Nexis Lens: Design Document

A gesture-augmented viewer for presenting code, git diffs and markdown. Keyboard and mouse do everything; hand gestures are an optional layer for presenting from across the room.

> Mock-ups are rendered from `design-document.html` (the HTML source) into `design/mockups/`. Gesture drawings are generated by `design/gestures.py`.

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2. Tasks

1. **Open a workspace:** choose a local folder (or reopen a recent one) so its files and git state appear in the sidebar.
2. **Browse and read files:** select files in the tree, scroll, search with Ctrl+F, and switch markdown between Raw and Preview.
3. **Review a git diff:** view working-tree changes or a past commit, step through hunks, and choose a unified or split layout.
4. **Switch tabs and use the session overview:** move between open items, or jump to any of them from a grid.
5. **Zoom and present:** scale content for a room and enter presentation mode with the interface hidden.
6. **Control the viewer with gestures:** turn on the camera, arm gestures, then scroll, zoom and flip modes on the focused file, and leave it for File mode to pick another file.

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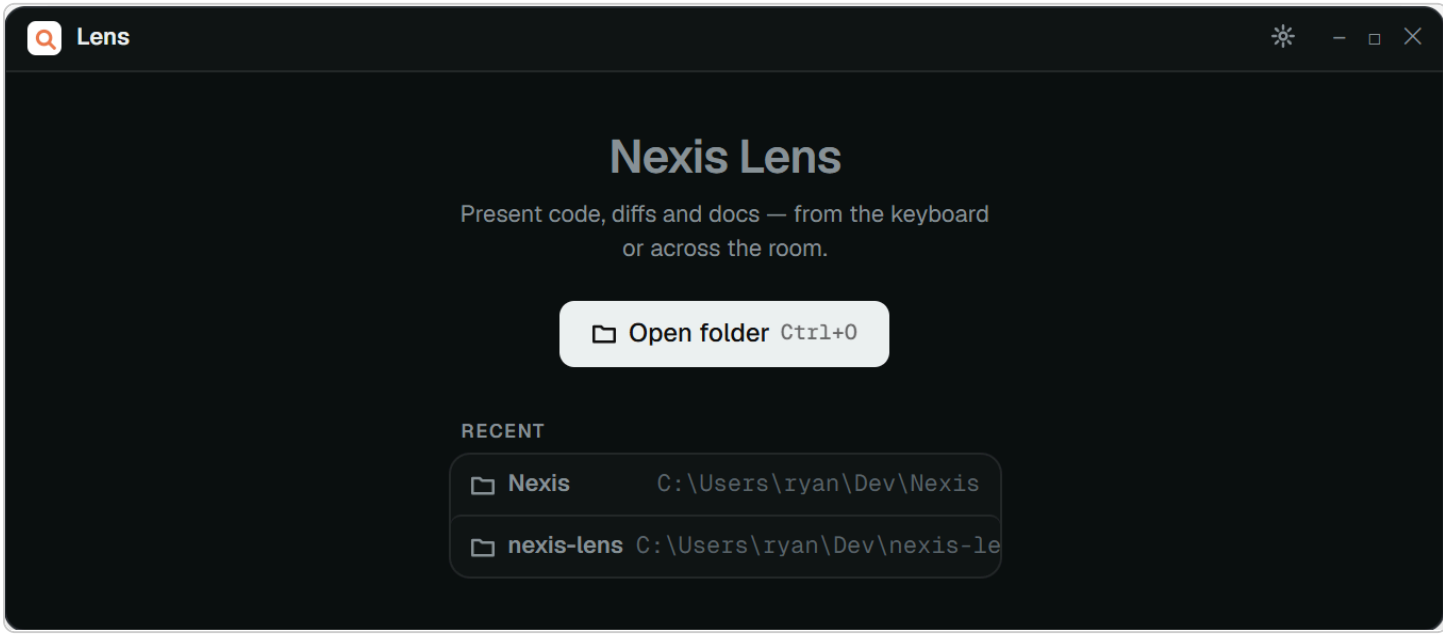
3. Prototypes

High-fidelity mock-ups drawn to match the running app (dark theme). Multi-step tasks are storyboards: read the frames top to bottom. Each is explained with Shneiderman's Eight Golden Rules.

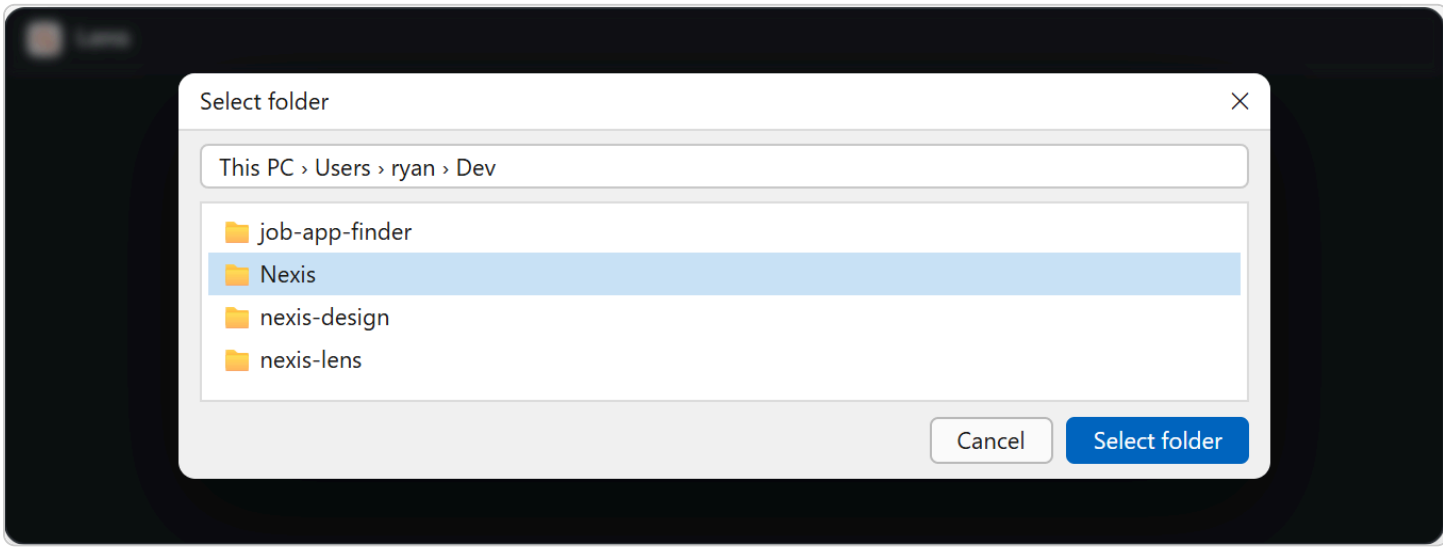
3.1 Open a workspace

Storyboard: 1a Welcome → 1b Pick folder → 1c Workspace loaded (or 1d Error).

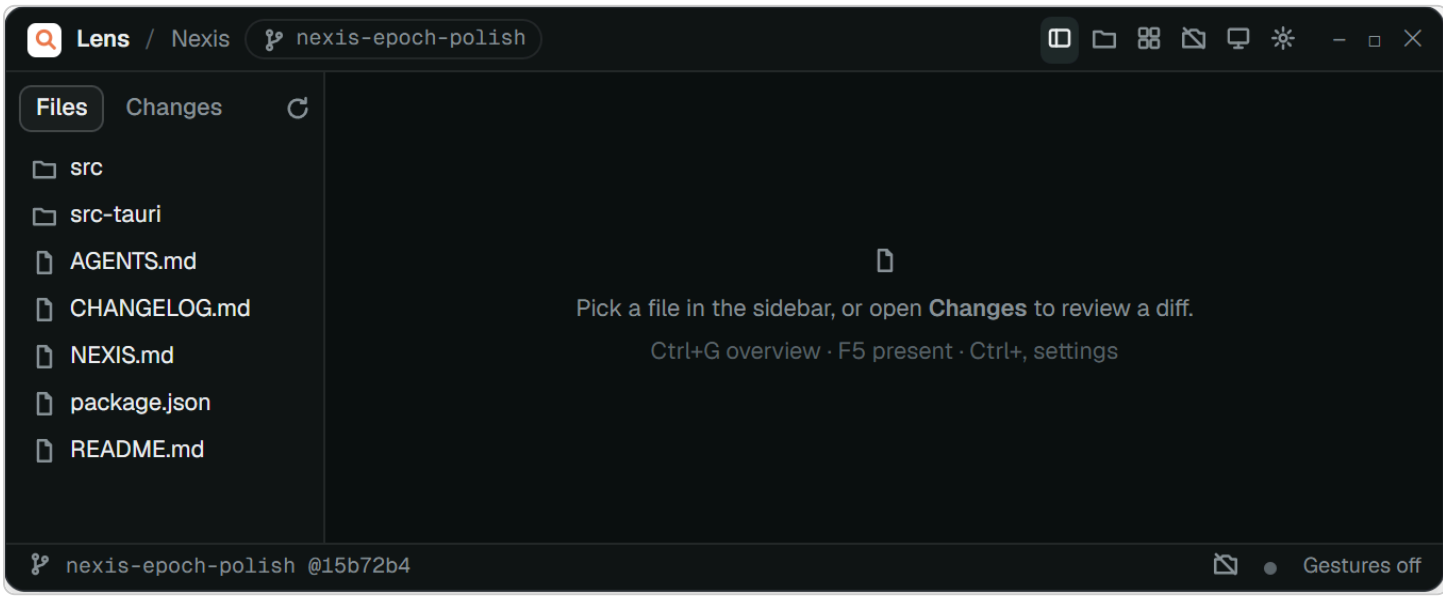
1a. Welcome screen (every launch starts here)



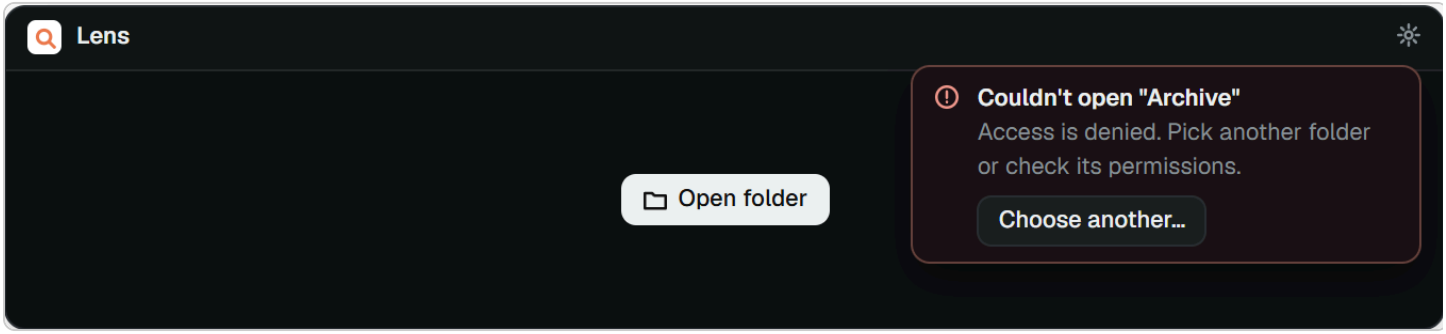
1b. Native folder picker



1c. Workspace loaded, nothing open yet



1d. Error: the folder could not be opened



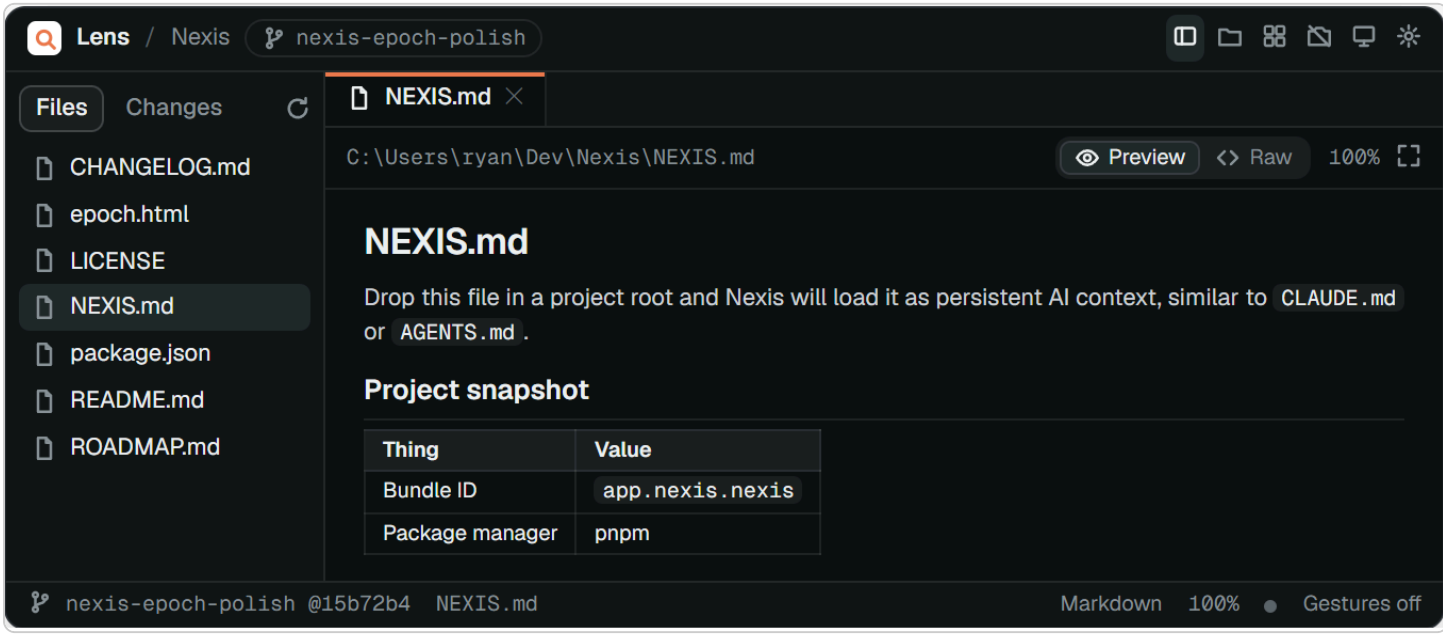
Eight Golden Rules

1. **Consistency:** the one primary action (Open folder) uses the filled button style used for primary actions everywhere.
2. **Shortcuts:** Ctrl+O opens the picker from anywhere; a recent workspace reopens in one click.
3. **Feedback:** once loaded, the title bar shows the workspace name and branch, and the status bar shows branch and commit.
4. **Closure:** the flow ends in a loaded workspace whose empty stage says what to do next.
5. **Error handling:** a failed open names the folder and the cause and offers "Choose another…".
6. **Reversal:** Cancel in the picker changes nothing; opening another folder is always one shortcut away.
7. **User control:** the user picks the folder; the app never scans the disk on its own.
8. **Memory load:** recents show name and full path, so users recognise rather than retype.

3.2 Browse files and read code or markdown

Storyboard: 2a Markdown in Preview → 2b Code with find.

2a. Markdown file in Preview mode



2b. Code file with Ctrl+F find (match 3 of 12)



Eight Golden Rules

1. **Consistency:** every file type shares the same tab bar, path bar and status bar; only the content area changes.
2. **Shortcuts:** Ctrl+F to find, Ctrl+Shift+V for Preview/Raw, Ctrl+= / Ctrl+− to zoom.
3. **Feedback:** the open file is highlighted in the tree and underlined in orange in the tab bar; find shows "3 of 12".
4. **Closure:** the Preview/Raw toggle shows the active mode the moment it changes.
5. **Error handling:** binary or unreadable files show a message in place of the content, not a blank pane.
6. **Reversal:** Esc closes find; the toggle flips back; closing a tab never touches the file.
7. **User control:** the viewer is read-only and never navigates on its own.
8. **Memory load:** the full path stays visible above the content, and per-tab scroll and zoom are restored.

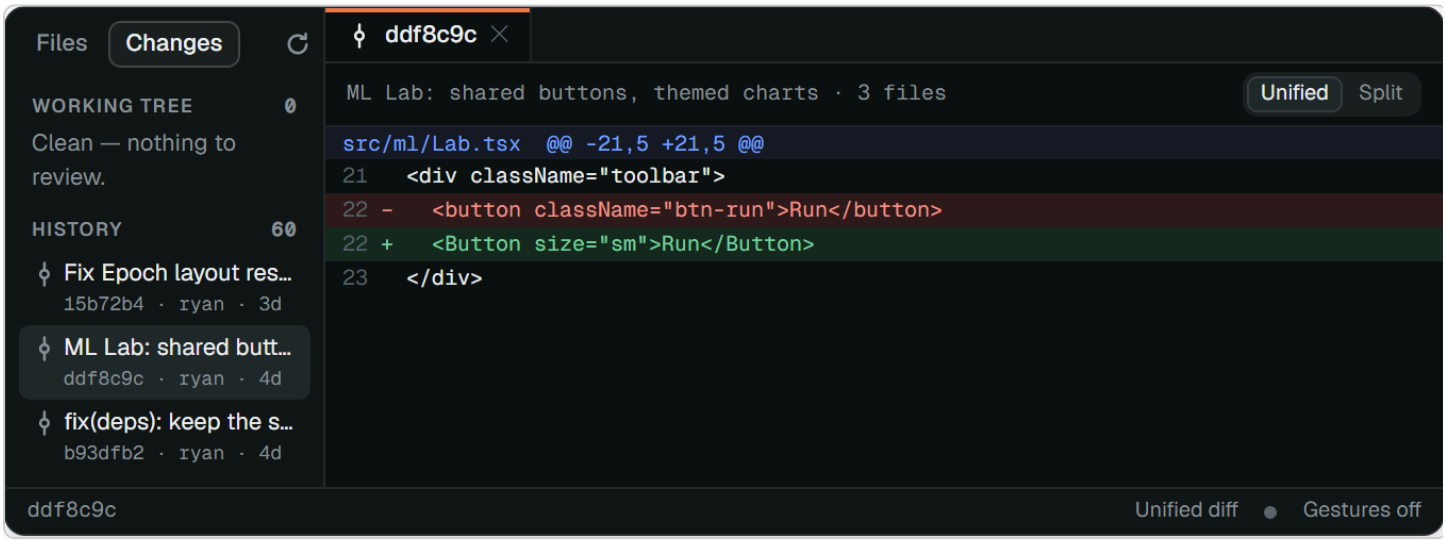
3.3 Review a git diff

Storyboard: 3a Working-tree change in split layout → 3b Past commit in unified layout. Both live in the Changes tab; history sits under the working tree.

3a. Changes tab: working tree, split diff, hunk 2 of 5



3b. A past commit from History, unified layout



Eight Golden Rules

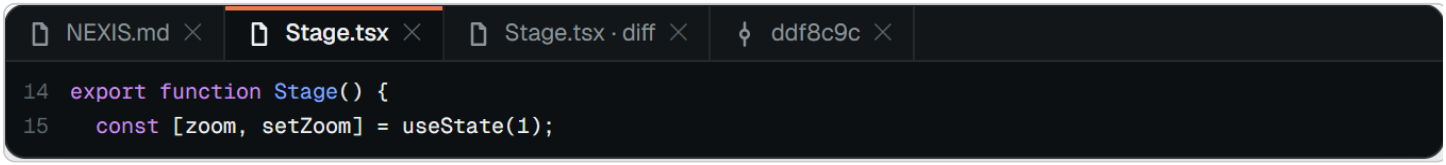
1. **Consistency:** added lines are green with "+", removed lines red with "−", in both layouts (never colour alone).
2. **Shortcuts:** `]` and `[` step through hunks; Ctrl+Shift+U switches unified/split.
3. **Feedback:** "Hunk 2/5", an orange marker on the current hunk, and +/− totals in the status bar.
4. **Closure:** stepping past the last hunk should say so instead of silently stopping (proposed; not built yet).
5. **Error handling:** a clean tree says "Clean — nothing to review"; a folder that isn't a repo says so.
6. **Reversal:** the view is read-only, so nothing here can damage the repo; the layout toggle flips back.

7. **User control:** the user chooses the file, the commit and the layout.
8. **Memory load:** the status bar names the layout and, when gestures are on, what the hand currently controls.

3.4 Switch tabs and use the session overview

Storyboard: 4a several tabs open → 4b Ctrl+G overview → Enter returns to the chosen tab.

4a. Tab bar with four open items (Ctrl+Tab cycles)



4b. Session overview: arrows move, Enter opens, Esc closes



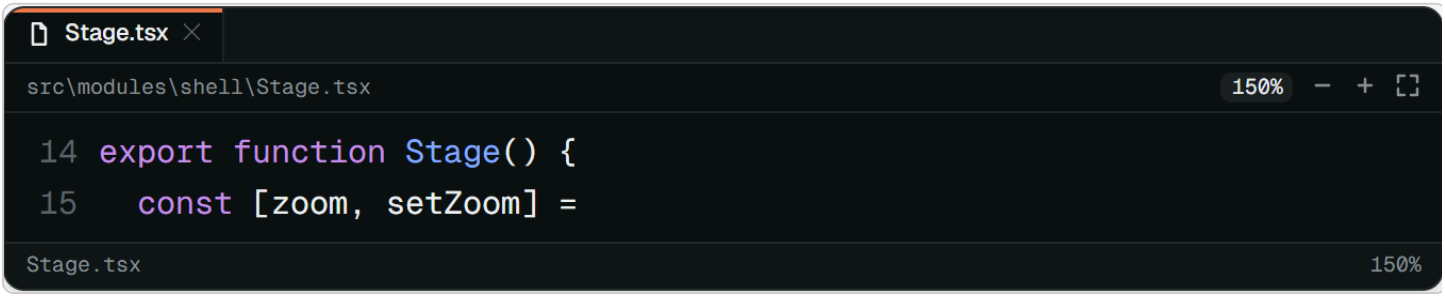
Eight Golden Rules

1. **Consistency:** the orange accent marks "current" in both the tab bar and the overview.
2. **Shortcuts:** Ctrl+Tab / Ctrl+Shift+Tab, Ctrl+G, arrow keys and Enter.
3. **Feedback:** the glowing card follows focus and carries a "current" badge.
4. **Closure:** Enter closes the overview on the chosen tab.
5. **Error handling:** with nothing open, the overview says so rather than showing an empty grid.
6. **Reversal:** Esc returns to the tab you came from.
7. **User control:** the overview opens only when asked for.
8. **Memory load:** each card previews its content and path, so tabs are recognised, not recalled.

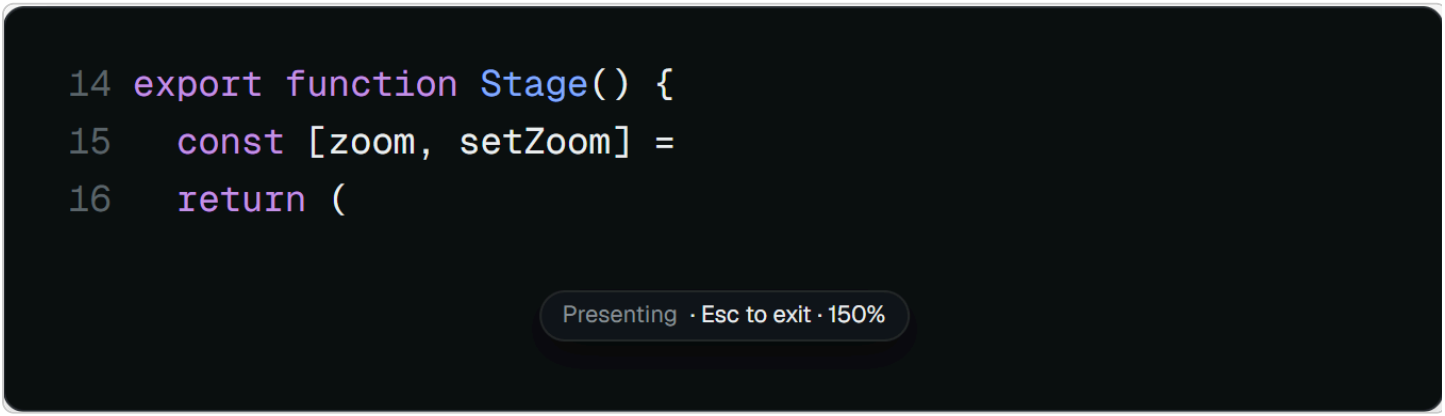
3.5 Zoom and present

Storyboard: 5a zoom for the room → 5b F5 presentation mode.

5a. Zoomed to 150% (Ctrl+=)



5b. Presentation mode (F5): no title bar, sidebar or status bar (exit hint is proposed, not built yet)



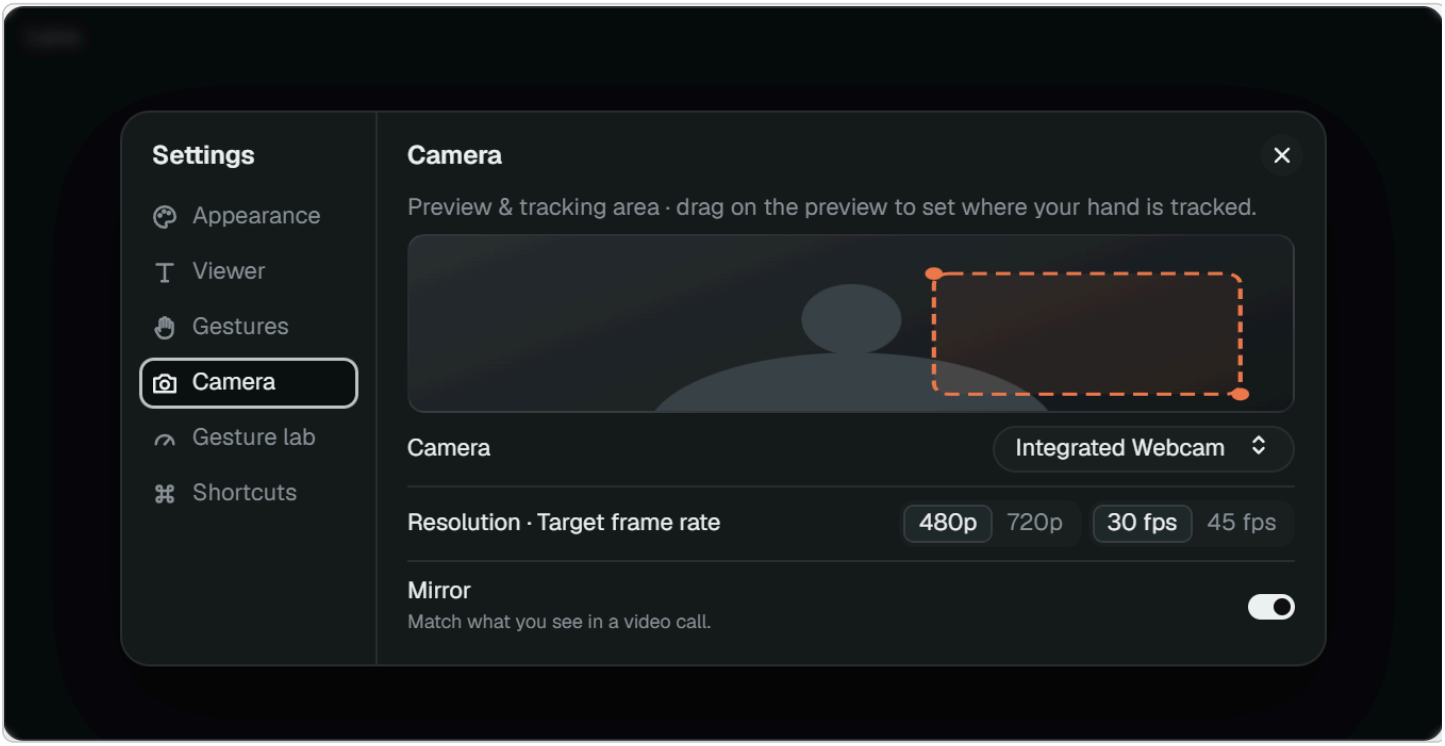
Eight Golden Rules

1. **Consistency:** the same zoom keys and the same zoom gesture work in code, markdown and diffs.
2. **Shortcuts:** F5, Ctrl+= / Ctrl+− / Ctrl+0, Ctrl+9 to fit.
3. **Feedback:** the zoom level appears next to the content and in the status bar.
4. **Closure:** entering and leaving presentation mode is one explicit action.
5. **Error handling:** zoom is clamped to a minimum and maximum so text never vanishes.
6. **Reversal:** Esc leaves presentation mode; Ctrl+0 resets zoom.
7. **User control:** nothing enters presentation mode on its own.
8. **Memory load:** a small "Esc to exit" hint stays visible while presenting.

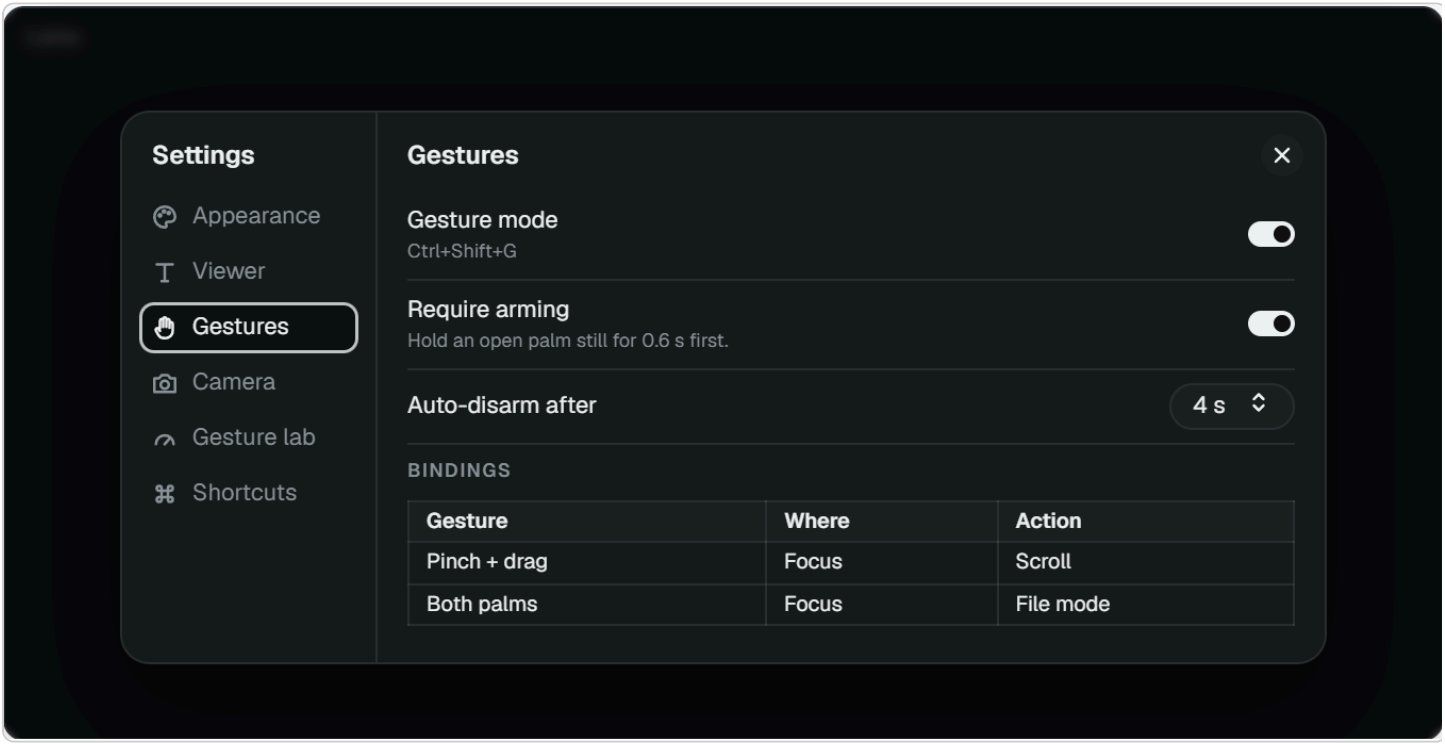
3.6 Control the viewer with gestures

Storyboard: 6a Settings → Camera → 6b Settings → Gestures → 6c Focus mode on a file → 6d File mode → (close hand) → back to 6c on the new file.

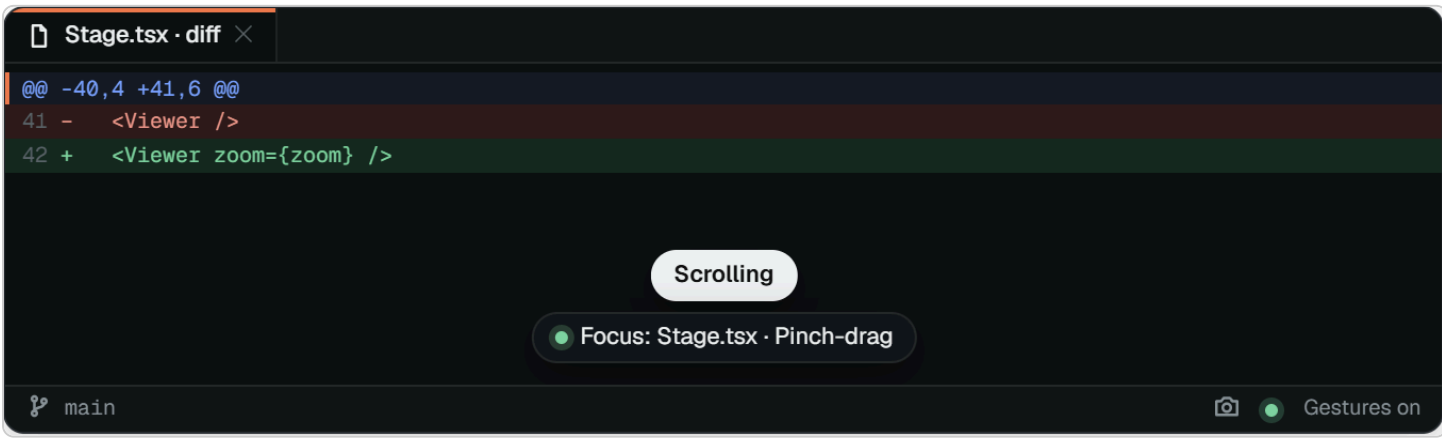
6a. Settings → Camera: preview and tracking area



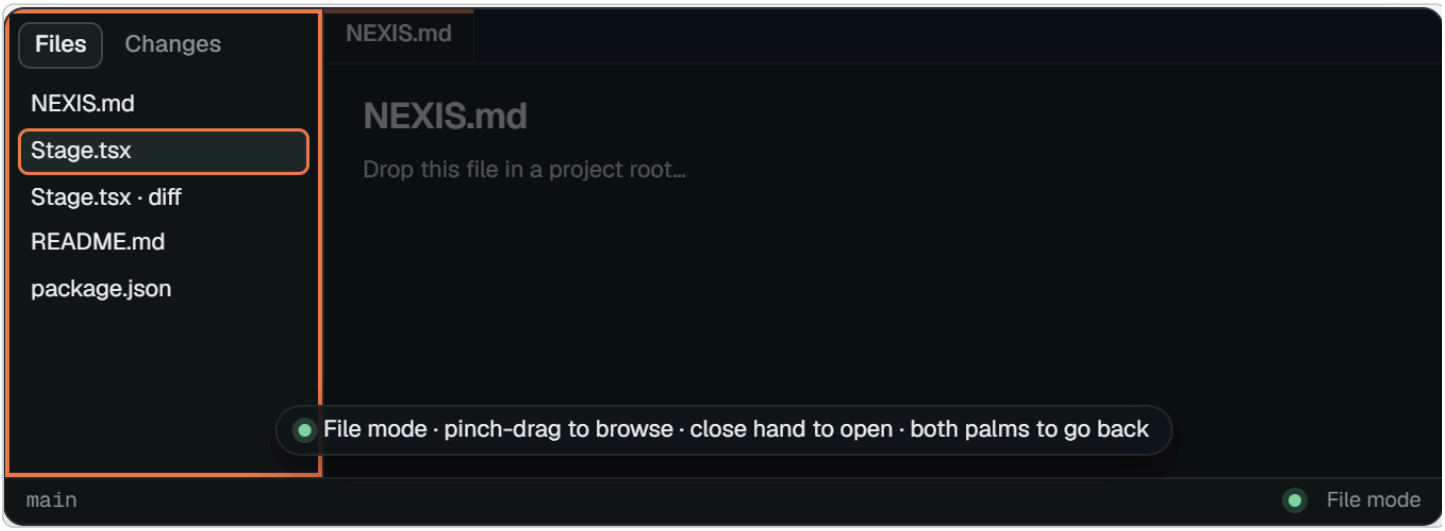
6b. Settings → Gestures: turn on, keep arming required, check the bindings



6c. Focus mode: gestures only act on the open file



6d. File mode (after showing both palms): gestures only move through the file list



Eight Golden Rules

1. **Consistency:** pinch-and-drag means "move this" in both modes: it scrolls the file in Focus mode and the list in File mode.
2. **Shortcuts:** Ctrl+, opens Settings; Ctrl+Shift+G switches gestures on or off without opening it.
3. **Feedback:** the HUD always names the mode and what the hand controls ("Focus: Stage.tsx"); File mode outlines the sidebar in orange.
4. **Closure:** closing the hand on a file opens it and drops back into Focus mode, which the HUD confirms.
5. **Error handling:** the HUD reports "Camera error", "No hand in frame" or "Low confidence" instead of guessing.
6. **Reversal:** showing both palms again in File mode cancels and returns to the file you left.
7. **User control:** gestures are opt-in and need a deliberate arm step; while a file is focused, no gesture can change files.
8. **Memory load:** only two modes, and File mode spells out its three gestures on screen.

3.7 Gesture design

The demo, and what it needs

The demo uses three items: a markdown file, a code file and a git diff. Each needs only a few controls:

Item	Needs
Markdown	Scroll, zoom, switch Raw ↔ Preview
Code	Scroll, zoom
Git diff	Scroll, zoom
Moving between them	Leave the file, pick another, open it

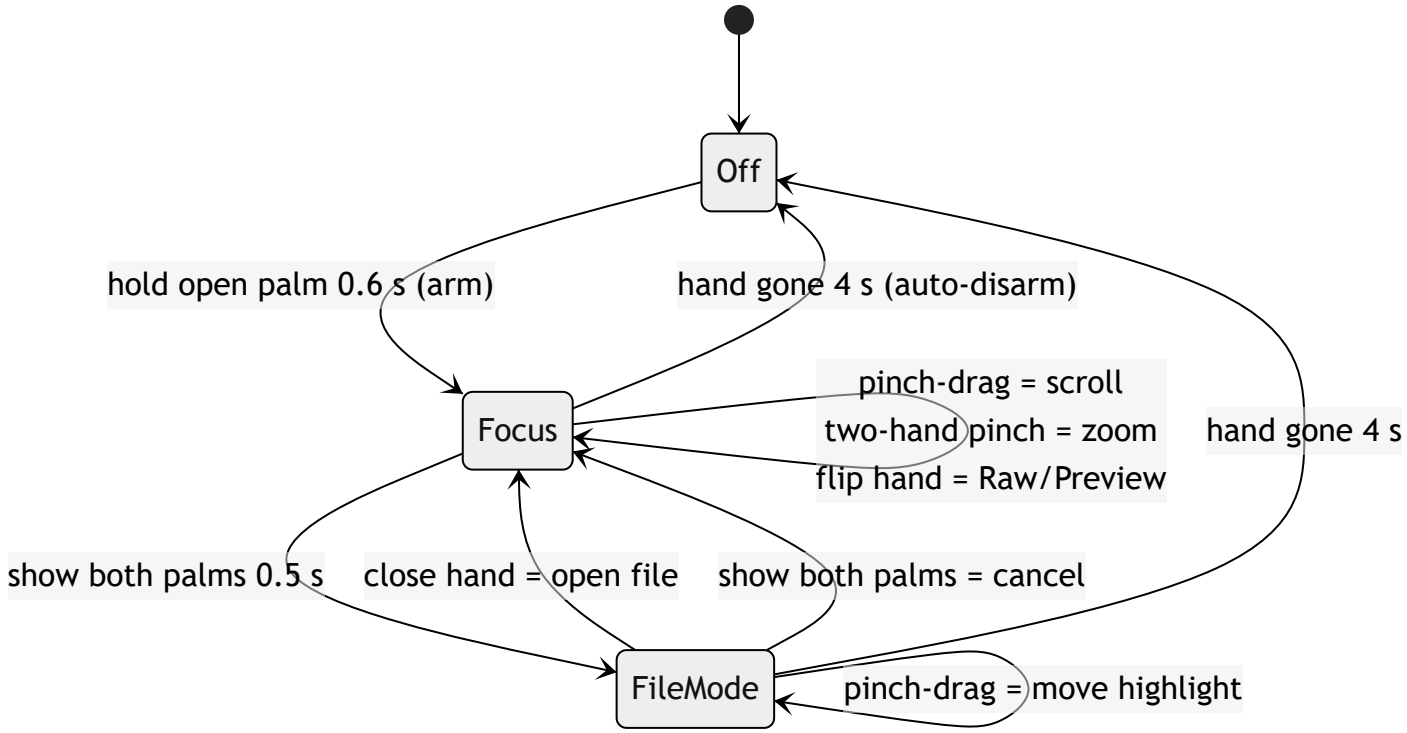
So the vocabulary can be small. A small set is easier to learn and gives the recogniser fewer poses to confuse.

The model: two locked modes

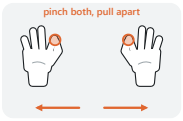
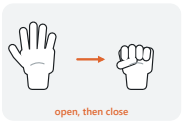
Gestures are **locked to one target at a time**, so a gesture can only ever do something that makes sense for what is on screen.

- **Focus mode:** the open file has the gestures. Hands can scroll, zoom and (in markdown) flip modes. No gesture can switch files, so a stray swipe while presenting cannot jump away.
- **File mode:** the file list has the gestures. Hands can move the highlight and open a file. Nothing scrolls or zooms the content.

A two-hand gesture is the only way between them, because two hands almost never happen by accident.



The proposed gesture set

	Gesture	Mode	Does	Keys
	**Hold an open palm** for 0.6 s	Off → Focus	Arms gestures	Ctrl+Shift+G
	**Pinch and drag** up or down	Focus	Scrolls the file, like dragging paper	wheel, arrows
	**Pinch with both hands, pull apart / push together**	Focus	Zooms in / out	Ctrl+= / Ctrl+–
	**Turn the hand over** (palm → back)	Focus, markdown only	Raw ↔ Preview	Ctrl+Shift+V
	**Show both open palms** for 0.5 s	Focus → File mode, or cancel	Leaves the file / goes back	Ctrl+G (overview)
	**Pinch and drag** up or down	File mode	Moves the highlight through the files	arrows
	**Open hand, then close it** (grab)	File mode → Focus	Opens the highlighted file	Enter

Why these gestures

- **Pinch-and-drag to scroll, not an open palm.** An open palm is how hands rest and wave while someone talks, so scrolling on any moving palm fires constantly. This is the Midas touch problem [1]. A pinch is deliberate: nothing moves until thumb and index touch, and letting go stops it, like a clutch. Apple's visionOS uses the same pinch-and-drag for scrolling [6], so many people will already know it.
- **Two-hand pinch to zoom.** "Spread to enlarge" is one of the most agreed-upon gestures in user-defined gesture studies [4]. A one-hand pinch can't zoom here because one-hand pinch now scrolls. A webcam also cannot measure depth well, so "push toward the screen" is not an option.
- **Turn the hand over for Raw / Preview.** Flipping matches the idea of seeing the other side of the same page. It only works in markdown, the only item with two modes, so it can't collide with anything else. It still has no study behind it, so it is the first candidate for the guessability test.
- **Both palms to leave a file.** Showing both open palms is the common "stop / hands off" sign. It needs two hands and a 0.5 s hold, which makes accidents very unlikely, and the hold separates it from two-hand zoom (which is pinched, not open). Surale et al. compared eleven bare-hand mode-switching techniques

[7]; using the non-dominant hand to switch modes is one of the options they tested. We have not checked their results in detail.

- **Grab (close the hand) to open.** Closing the hand reads as "take this one". It is a distinct pose that MediaPipe already detects reliably (the fist classifier exists). Point-and-dwell selection is more accurate but users preferred an active gesture in one study [8]; grab is an active gesture.
- **Swipes are dropped from the demo.** Swipes were the main way to change tabs, which is exactly what Focus mode forbids. Fast flicks also misfire on the return stroke. Hunk stepping in the diff stays on `]` and `[`.
- **Keep the arming step.** Arming separates "I'm presenting with my hands" from "I'm just talking". Auto-disarm after 4 s with no hand keeps it safe if the presenter walks away.

How each gesture is recognised

MediaPipe returns 21 landmarks per hand. `recognizer.ts` sorts each frame into one of five poses (open, fist, point, pinch, other) from finger geometry, then turns pose plus motion into events. Distances are divided by palm size (wrist to middle knuckle), so thresholds hold near or far from the camera.

Gesture	Detected when	Status in code
Open palm (pose)	At least 4 fingers extended (tip more than 1.15× further from the wrist than its middle joint)	Exists
Arm	Open palm moving under 0.18 frame-widths/s for 600 ms	Exists
Pinch (pose)	Thumb–index gap under 0.28 palm sizes; released above 0.40 (the gap stops flicker)	Exists
Pinch-drag	While pinched, palm movement above a small dead zone becomes scroll	Built
Two-hand pinch zoom	Both hands pinched; the change in distance between the pinch points sets the zoom	Built
Flip	Open palm whose facing (sign of the palm normal from wrist and two knuckles) changes within 700 ms	Exists
Both palms	Two open hands, both nearly still, for 500 ms	Built
Grab	Open pose followed by fist within ~500 ms, held 140 ms	Built
Mode lock	Dispatcher only resolves bindings for the current mode	Built (`bindings.ts`, `picker.ts`)

Only the highest-confidence hand drives one-hand gestures, so two hands never fire twice. The dispatcher adds a confidence floor, the arming gate, auto-disarm after 4 s and a 450 ms cooldown between discrete commands.

How we will validate the gesture set

- **Guessability test** [4]: show 5 to 8 people each outcome ("scroll down", "zoom in", "show the raw markdown", "go to another file", "open this file") and record the gesture they try first. Keep a gesture if most people agree on it; flip is the most at risk.
- **Accidental-command count:** five minutes of normal presenting in Focus mode, counting any command nobody meant. The goal is zero file changes and very few stray scrolls.

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4. Non-interface components

- **Shell and backend:** Tauri v2 app (Rust backend, React 19 + TypeScript front end). Typed IPC connects the two; a browser-only mode serves an in-memory demo workspace.
- **Files:** the Rust backend opens folders, lists files respecting .gitignore, and reads file contents on request.
- **Git:** the backend runs the git CLI for status, diff, log and show; the front end parses unified diffs for the unified and split views.
- **Command bus:** keyboard, toolbar and gestures all call `runCommand(id, args, source)`; the active view handles it. This is why every task works without a camera.
- **Gesture pipeline:** camera → MediaPipe HandLandmarker (WASM and model bundled under `public/mediapipe/`, no CDN, works offline) → recogniser (landmarks to poses to events) → dispatcher (confidence floor, arming, mode lock, auto-disarm, cooldown) → bindings → command bus.
- **Performance plan:** target at least 30 fps on an 8 GB integrated-GPU laptop. If short, drop to 480p, skip frames, or move inference to a worker with OffscreenCanvas.
- **Privacy and permissions:** camera video is processed on-device and never uploaded; macOS builds include the camera entitlement and usage description.
- **Distribution:** GitHub Actions builds Windows (NSIS/MSI) and macOS (universal dmg) installers from version tags.

References

1. *Designing Gunslinger: An Intermodal Large Display Interaction*, University of Waterloo. Arms-down stance, Midas touch, hand-cursor feedback. https://uwspace.uwaterloo.ca/handle/10012/9358
2. arXiv 2603.15991. Gorilla arm and Midas touch as barriers to gesture and gaze interfaces. https://arxiv.org/pdf/2603.15991
3. US patent 11301049, *User interface control based on elbow-anchored arm gestures*. https://patents.google.com/patent/US11301049
4. Wobbrock, Morris, Wilson. *User-defined gestures for surface computing*, CHI 2009 (guessability method). https://microsoft.com/en-us/research/wp-content/uploads/2009/04/SurfaceGestures_CHI2009.pdf
5. *MotionInput v2.0: gesture-based methods for controlling software with a webcam*, arXiv 2108.04357. https://arxiv.org/pdf/2108.04357

6. Apple Human Interface Guidelines, *Gestures* (visionOS: look and pinch to select, pinch and drag to scroll). Seen through third-party summaries only; check against developer.apple.com before citing. https://developer.apple.com/design/human-interface-guidelines/gestures
7. Surale, Matulic, Vogel. *Experimental Analysis of Barehand Mid-air Mode-Switching Techniques in Virtual Reality*, CHI 2019; and Surale's thesis *Barehand Mode Switching in Touch and Mid-Air Interfaces*. https://uwspace.uwaterloo.ca/handle/10012/15950
8. Point-and-dwell vs. push selection on large displays (10 participants). https://doi.org/10.1145/2838739.2838819
9. *Eliciting mid-air gestures for wall-display interaction*, NordiCHI 2016: elicited gestures were larger and more physical than on small screens. https://doi.org/10.1145/2971485.2971503

Research caveats: the gesture choices rest on a handful of web searches, not a literature review. Refs 6–9 were found through abstracts and summaries and have not been read in full. Wobbrock [4] studied touch surfaces, so it supports the metaphors, not mid-air performance. Flip and both-palms have no direct study behind them; the guessability test decides.